



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 2 • STANDARD 6.DS.4

Describe Data Using the Mean

Lesson 2-8 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can determine the mean of a data set, use a target mean to find a missing value, and explain the mean as the fair share and the balance point of the data.

Language Objective: I can explain my reasoning using the words mean, average, measure of center, balance point, and data set.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mean

Average

Measure of center

Balance point

Data set



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The value found by adding all the data and dividing by how many values there are is the .

2

A common everyday word that means the same thing as the mean is the .

3

A single number that describes the typical value of a whole data set is a .

4

The value where the total amount above equals the total amount below is the .

5

The whole collection of collected values is the .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

COMPARE

How many points does Sofia need to score in the last game to hit her goal, and how does the balance-point view give the same answer as adding and dividing?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Mean** — The average of a data set. Add all the values, then divide by how many values there are.
Media — El promedio de un conjunto de datos. Suma todos los valores y divide entre cuántos valores hay.

- ☐ **Average** — Another word for the mean — the amount each value would be if the total were shared equally.
Promedio — Otra palabra para la media: la cantidad que tendría cada valor si el total se repartiera en partes iguales.

- ☐ **Measure of center** — A single number used to describe the middle, or typical value, of a whole data set.
Medida de centro — Un solo número que se usa para describir el centro, o valor típico, de todo un conjunto de datos.

- ☐ **Balance point** — The value where the total distance of the data above it equals the total distance below it. The mean is the balance point.
Punto de equilibrio — El valor donde la distancia total de los datos por encima es igual a la distancia total por debajo. La media es el punto de equilibrio.

- ☐ **Data set** — The whole collection of values that were gathered or measured.
Conjunto de datos — Toda la colección de valores que se recopilaron o midieron.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

After 9 games Sofia has scored ____ points. To average 15 across 10 games she needs ____ points in all, so game 10 must be ____ points. Using the balance point instead, her games total ____ points below the target mean and ____ points above it, so game 10 has to land ____ the target mean.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A typical day is the amount Miguel would earn if he earned the SAME amount every day.

Evidence: Students say a typical day is the equal-share amount, and that sharing \$60 across 4 days means dividing.

Reasoning: This evidence connects the definition of mean to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- After 9 games Sofia has scored ____ points. To average 15 across 10 games she needs ____ points in all, so game 10 must be ____ points. Using the balance point instead, her games total ____ points below the target mean and ____ points above it, so game 10 has to land ____ the target mean.

- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.

Mi afirmación es ____.

- I know because ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
 - **My math evidence is ____.**
Mi evidencia matemática es ____.
 - **This proves ____ because ____.**
Esto demuestra ____ porque ____.
-
-
-

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Mean
2. **Notes 2:** Average
3. **Notes 3:** Measure of center
4. **Notes 4:** Balance point
5. **Notes 5:** Data set
6. **Watch for this mistake:** *The most common mistake with the mean is adding correctly and then dividing by the wrong number — or not dividing at all. Students report the total (for 8, 10, 6, 12 they say the mean is 36) or they divide by a count that does not match the data, such as dividing a 4-value total by 5, or skipping a zero when counting the values. Before you finish, do two things: count the values in the data set out loud, and check that your mean lands between the smallest and largest value. A mean of 36 for data that never goes above 12 is impossible.*
7. **Try It 1:** A. 17 points — *She needs 150 points in all and already has 133, so game 10 must supply $150 - 133 = 17$ points.*
8. **Try It 2:** A. 24 — *If every value moves up by 4, the balance point moves up by 4 as well, so the new mean is $20 + 4 = 24$. In totals: adding 4 to each of n values adds $4n$ to the total, and $4n \div n = 4$.*